which is a constant (say k). Now let us find the ratio for each of the value from the table.

$\frac{x}{y} = \frac{1}{500} = \frac{2}{1000} = \frac{3}{1500} = \frac{6}{3000} = \frac{7}{3500}$ and so on. All the ratios are equivalent and its simplified form is $\frac{1}{500}$.

PBEK1P

In general, $\frac{x}{y} = \frac{1}{500} = k$ (k is a constant)

When x and y are in direct proportion, we get $\frac{x}{y} = k$ or $x = ky$ (k is a constant)

Considering any two ratios given above, say, $\frac{2}{1000} = \frac{6}{3000}$, where 2($x_1$) and 6 ($x_2$) are the number of shirts (x) and 1000 (y_1) and 3000 (y_2) are their prices (y). So, when x and y are in direct proportion we can write $\frac{x_1}{y_1} = \frac{x_2}{y_2}$ [where, y_1, y_2 are values of y corresponding to the values x_1, x_2 of x].

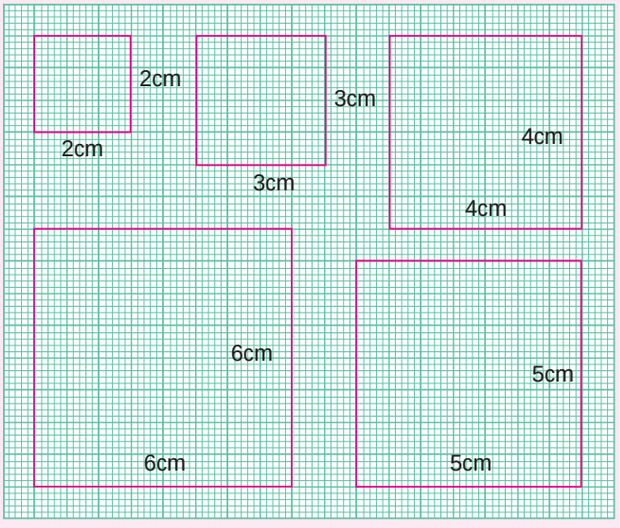
Think

The number of chocolates to be distributed to the number of children. Is this statement in direct proportion?

Try this

Observe the following 5 squares of different sides given in the graph sheet

The measures of the sides are recorded in the table given below. Find the corresponding perimeter and the ratios of each of these with the sides given and complete the table.



Side of the square (x) in cm	2	3	4	5	6
Perimeter of the square (y) in cm					
$\frac{x}{y}$					

From the information so obtained state whether the side of a square is in direct proportion to the perimeter of the square

7th_Maths_T1_EM_Chp4.indd 74 17-12-2019 17:17:23